ABSTRACT

Construction sites pose significant safety risks, often due to non-compliance with personal protective equipment (PPE) requirements. Traditional manual supervision methods are inefficient and prone to errors. This research presents the Construction Site Safety Analyzer, an AI-driven system that automates PPE detection using YOLOv8 and Region-Based CNN (R-CNN). The system analyses live video feeds to identify workers and their protective gear, ensuring real-time compliance monitoring. When a violation is detected, a GSM module sends immediate alerts to supervisors for prompt action. By leveraging deep learning and IoT, the system enhances workplace safety while reducing human oversight errors.

Designed for adaptability, it can be deployed on edge devices or cloud-based platforms, making it suitable for diverse construction environments. Performance is evaluated using precision, recall, and F1-score metrics, demonstrating high accuracy in detecting PPE violations. Experimental results confirm that the system effectively minimizes manual supervision efforts, strengthens safety enforcement, and mitigates accident risks. Future improvements may include enhanced detection in low-light conditions and faster processing through edge computing. This approach provides a smart, automated solution for improving construction site safety.